M10 — What this block gets you

- The governance tooling behind your single source of truth— Knowledge Catalog, Sensitive Data Protection, fine-grained security
- The moment: Google's own Bronze/Silver/Gold zone slides held up next to your architecture doc— zone for zone
- A GCP home for each zone— GCS/BigLake Bronze, Dataform Silver-to-Gold, native BigQuery Gold marts
- A phased migration path off the SQL Server DW— no big-bang cutover, extracts retired last
- The cost levers and best practices you set on day one— storage class, bytes scanned, budgets and alerts

M10 — 14:00 to 14:35, then break

14:00–14:10	Governance as your single-source-of-truth requirement — Knowledge Catalog, Sensitive Data Protection
14:10–14:15	Fine-grained security — column-level, row-level, dynamic masking
14:15–14:25	The Medallion — Google's Bronze/Silver/Gold zones next to your architecture doc
14:25–14:35	Migration strategy, cost management, best practices
14:35–14:50	Break
14:50–15:20	M11 — the BQML direction and the Power BI last mile

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What is metadata?

Metadata is data about data, which can include:

01 Who created the data

02 When it was created

03 What it contains

04 How it relates to other data

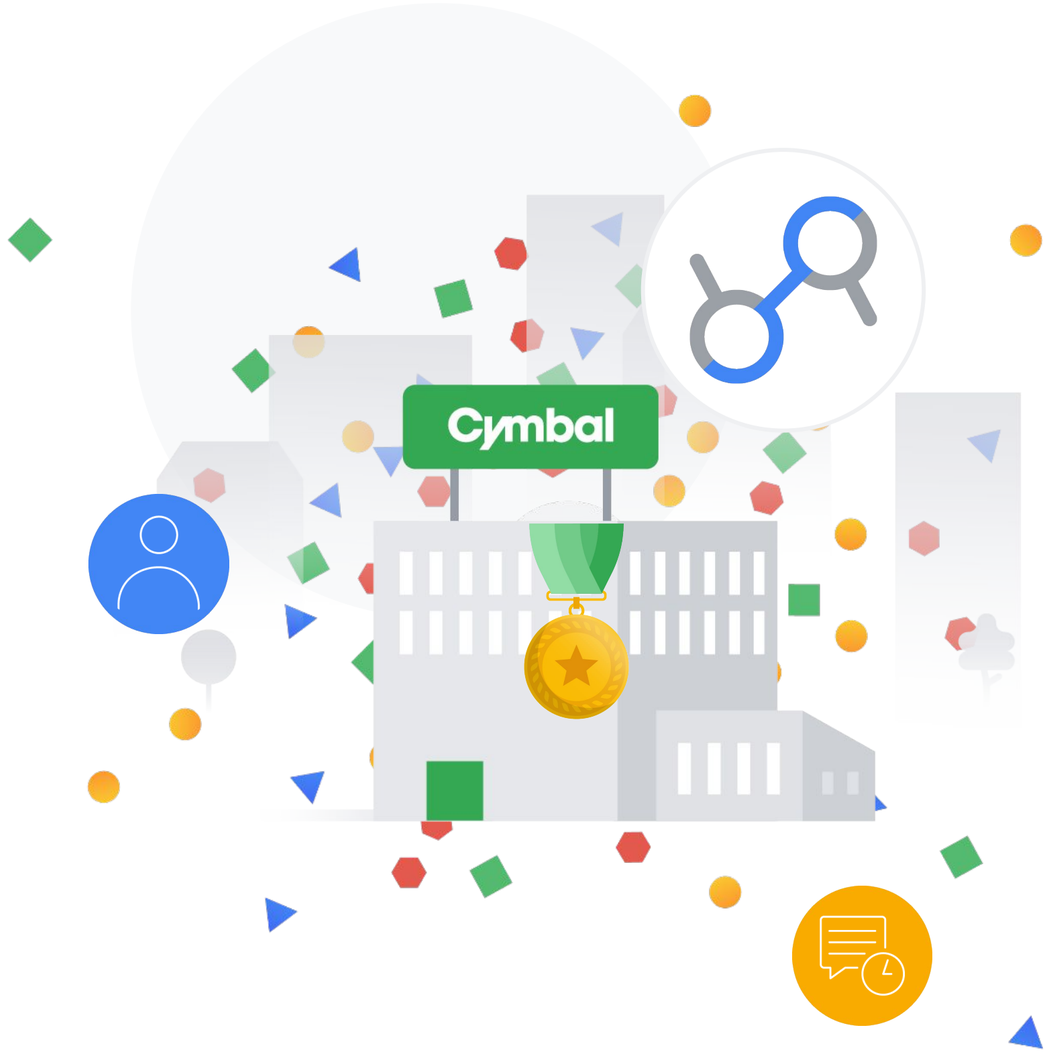
05 Who owns it

06 Its security sensitivity

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Knowledge Catalog for Cymbal

Without a centralized metadata system, data management can be difficult. Knowledge Catalog provides a unified metadata hub.

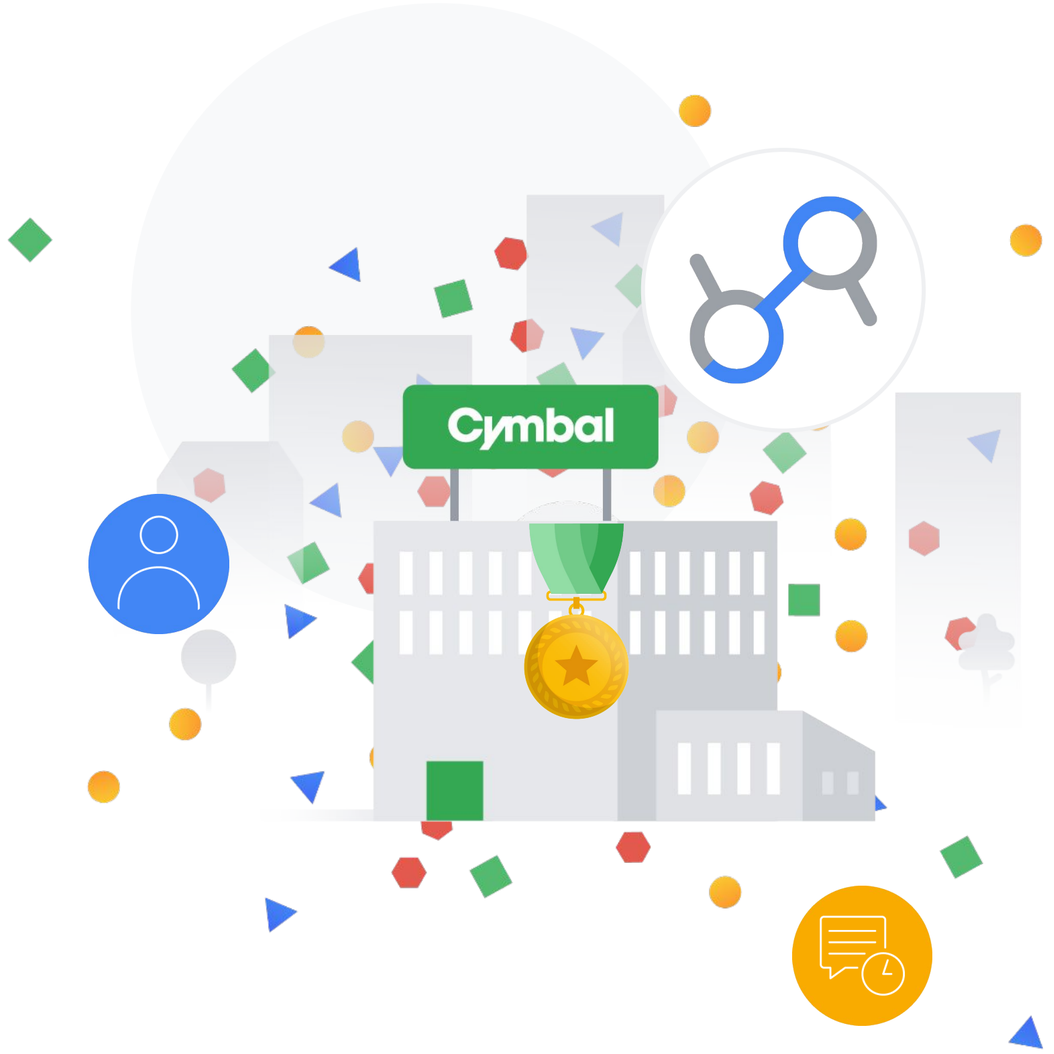


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Knowledge Catalog for Cymbal

01 Acts as a universal catalog for all data assets, whether they reside in BigQuery, Cloud Storage, or Lakehouse.

02 Use to discover data, understand data lineage, and manage and augment metadata.



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By providing a single reference source, Knowledge Catalog helps Cymbal manage data at scale while ensuring consistency and quality.

Governance = your single source of truth

Your problem is not storing data — extracts scatter it. One governed copy serves everyone.

- Knowledge Catalog— one catalog across BigQuery, Cloud Storage, lakehouse; which table is real has an answer
- Sensitive Data Protection— finds and classifies PII before it reaches a dashboard; your Data Discovery analogue
- Column-level security + masking— purchase history yes, contact info no; your column GRANT/DENY plus masking
- Row access policies— the regional manager sees only their region, like your RLS predicates
- Lineage— change a Silver table, see exactly which Gold marts and Power BI reports you touched

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Sensitive Data Protection



Protect customers' sensitive information, such as names, addresses, and credit card numbers.



A data breach can damage Cymbal's reputation and lead to significant financial penalties.



Sensitive Data Protection is an essential tool for securing data.

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Sensitive Data Protection

Discover

Run scans on data lakehouse and other data to identify multiple types of sensitive field.

Classification

Data is classified by sensitivity level. This ensures the right security controls are applied.

Protection

Use techniques, such as masking or tokenization, to de-identify the data.

Example: ******9787

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This proactive approach to data protection allows Cymbal to use their data for analytics and ML without exposing sensitive information.

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Fine-grained security

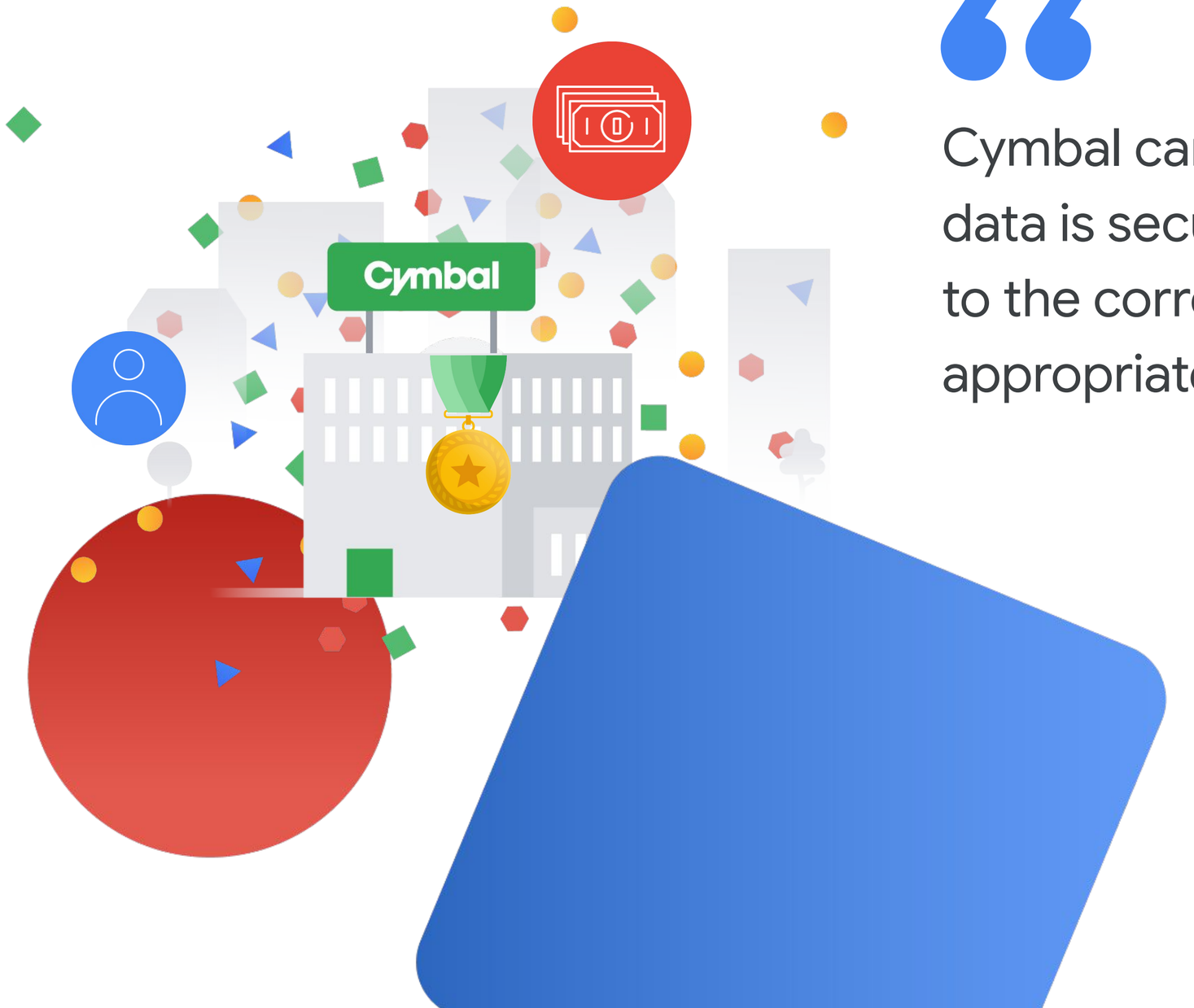
Column-level security

Restricts access to specific columns in a table. For example, a marketing analyst might be able to access a customer's purchase history but not their contact information. This is effective for protecting Personally Identifiable Information (PII).

Row-level security

Filters which rows a user can access. A regional sales manager for North America, for instance, would only have access to sales data for that region. This is particularly useful for large, multinational companies, such as Cymbal.

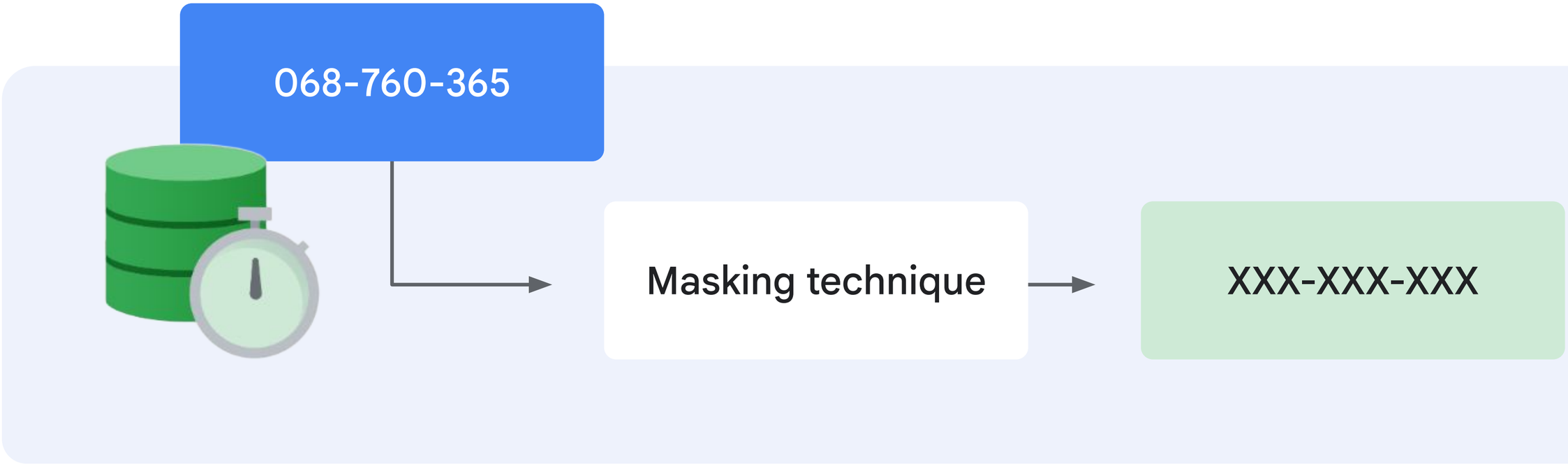
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Cymbal can ensure that their data is secure and accessible to the correct people at the appropriate time.

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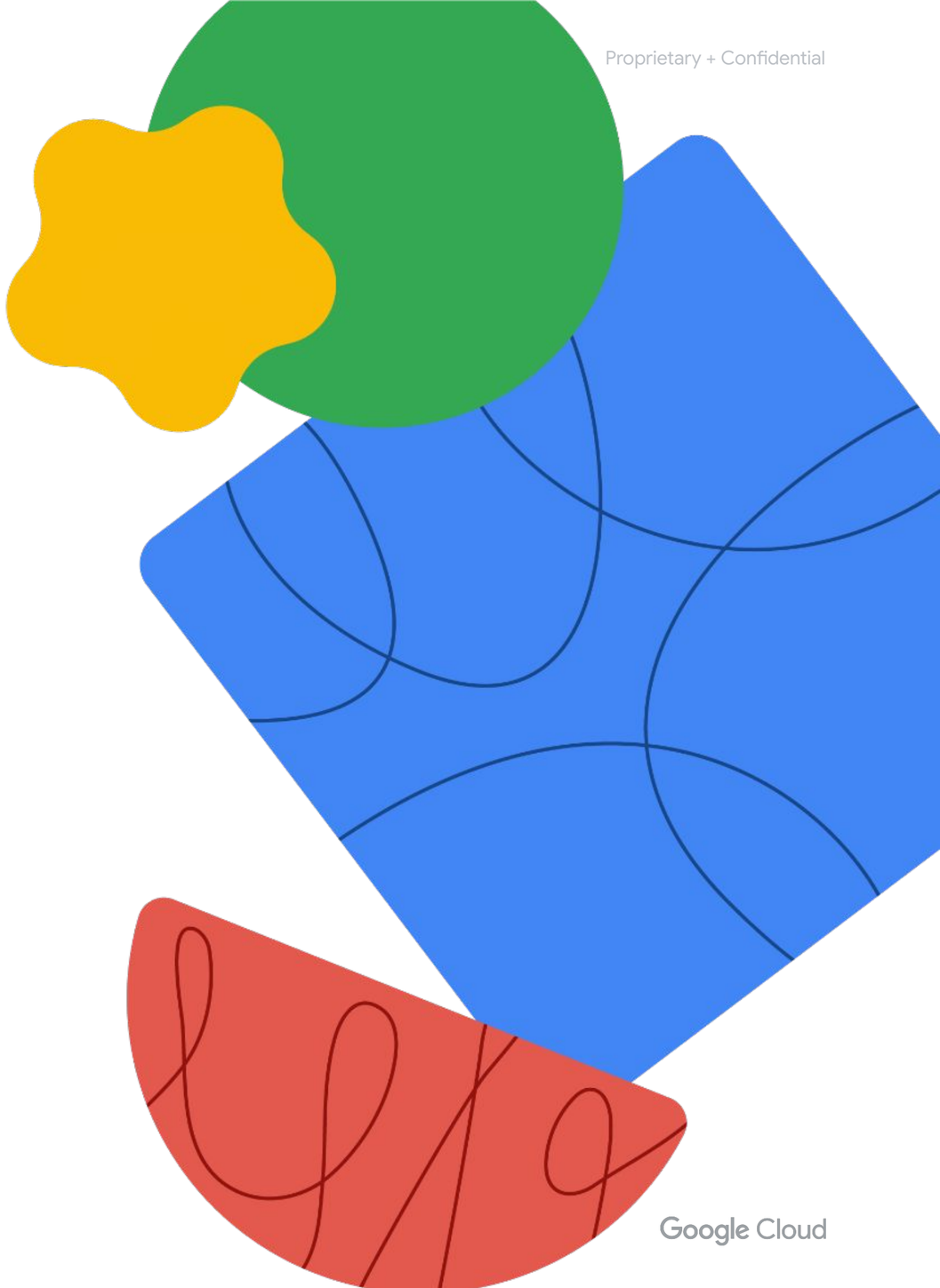
For Lakehouse tables in Cloud Storage, Sensitive Data Protection can apply dynamic data masking.



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Data loss prevention

Demonstration



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Scenario: Cymbal loyalty program

01

New customer information has been collected and loaded into a BigQuery table named `loyalty_program_customers`.

02

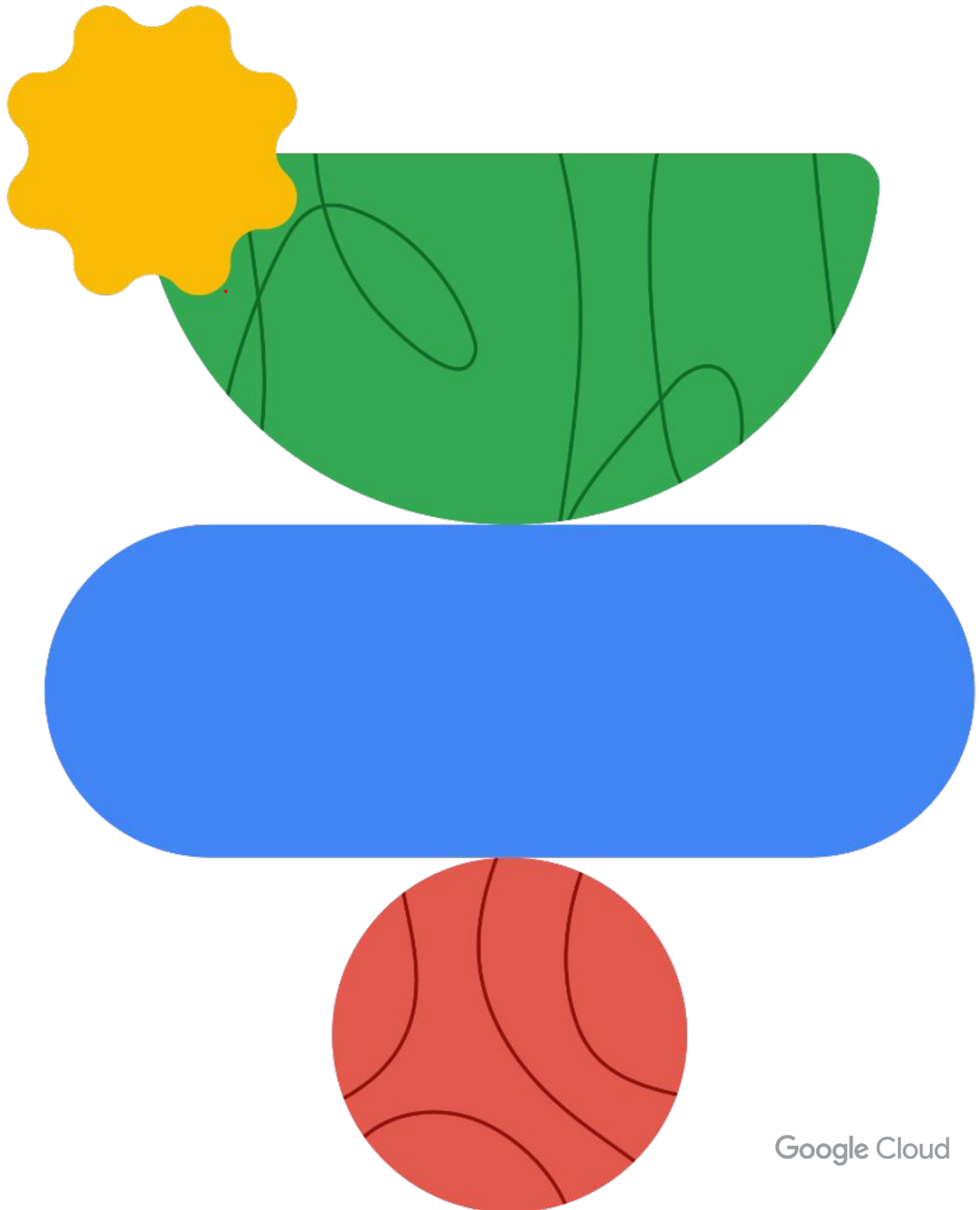
This table contains standard information, such as names and purchase histories.

03

Includes potentially sensitive data, such as email addresses, phone numbers, and free-text comments from customer feedback surveys.

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Before this data is made available to the marketing analytics team, any PII must be properly handled to avoid exposing sensitive customer details in analytics dashboards.



Your architecture doc, in Google's deck

Hold the next six slides up next to the Iowa architecture doc — they line up zone for zone.

- Retire legacy extracts— the nightly SSIS extract jobs end; data lands in BigQuery continuously
- SQL Server DW to BigQuery— one governed copy becomes the analytics source of truth
- Bronze, Silver, Gold— your medallion lakehouse, in Google's own vocabulary
- Dataform ELT Silver to Gold— versioned, tested SQL pipelines replacing SSIS data flows
- Gold marts feed Power BI— the reports your users already have, repointed at BigQuery

Your design, in their deck.

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The medallion architecture (1 of 3)

Bronze zone

This is the landing zone for all raw data.

Silver zone

In this zone, initial transformations happen.

Gold zone

Data is aggregated and optimized for analytics and reporting.

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Cymbal: Bronze zone (raw data)

01

Clickstream data from their website.

02

Batch exports of transactional data from their e-commerce database.

03

JSON data from their social media marketing campaigns.

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Cymbal: Silver zone (cleansed and conformed data)

01

Parsed clickstream data with structured sessions.

02

Augmented transactional data.

03

Standardized date and time formats.

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Cymbal: Gold zone (curated business-level data)

01

Aggregated daily sales tables.

02

Customer 360-degree view tables.

03

Inventory performance summaries.

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The medallion architecture (2 of 3)

Bronze zone

A historical record of what was received.

Silver zone

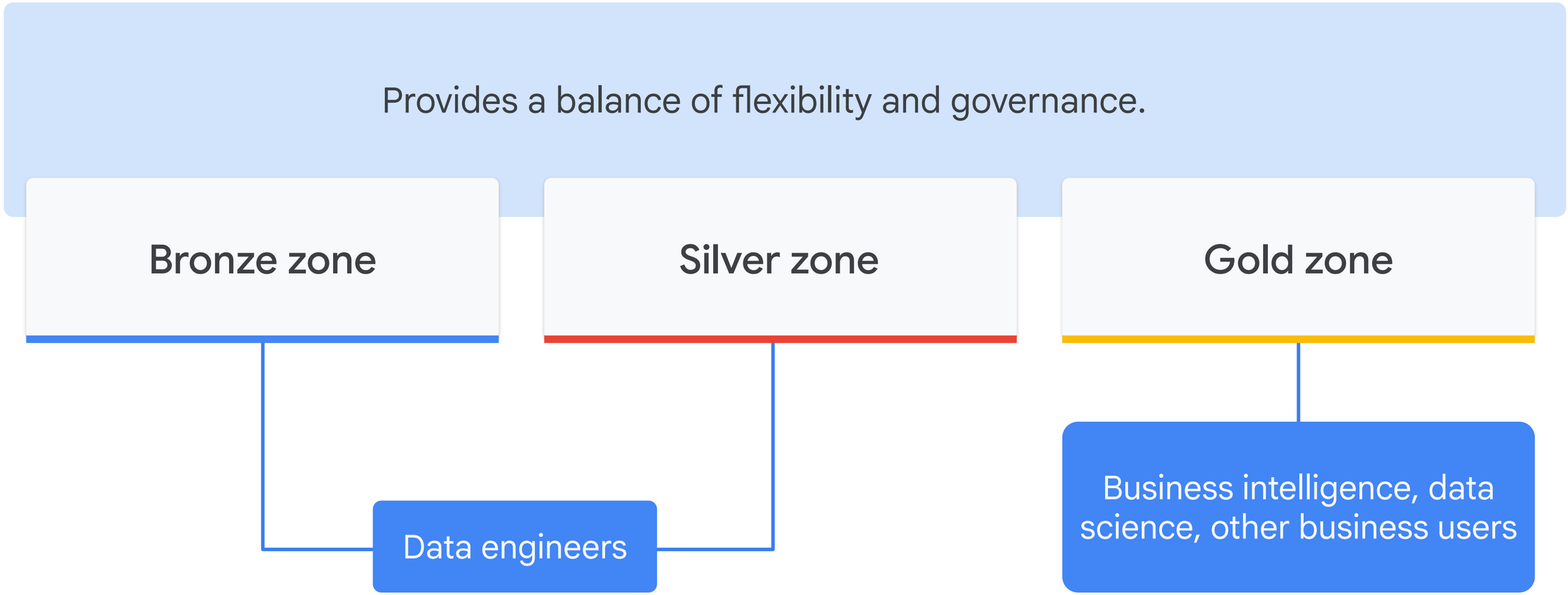
Data often stored in an open format in Cloud Storage but queryable with BigQuery.

Gold zone

Normally native BigQuery tables for maximum query performance.

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The medallion architecture (3 of 3)

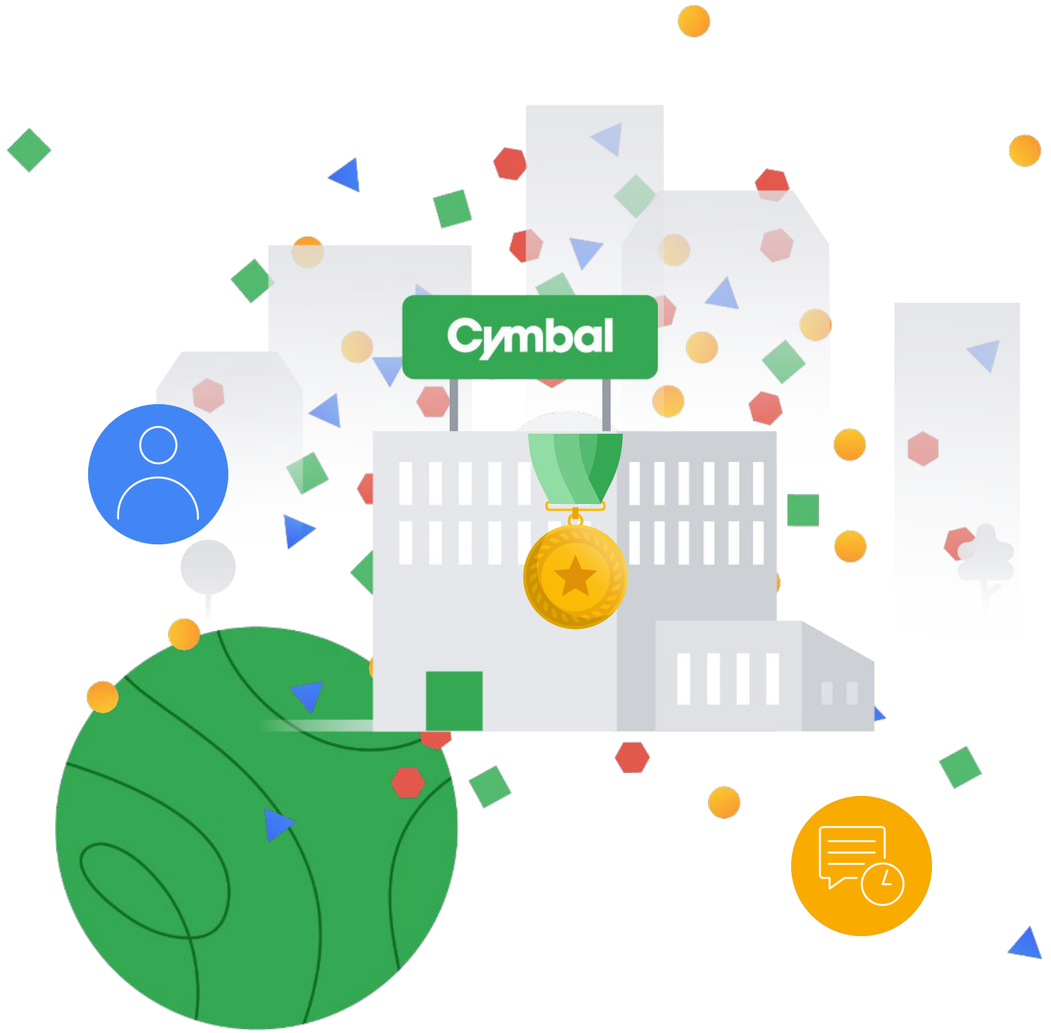


Where each of your zones lives in GCP

Same zones as your doc — here is the service behind each box.

- Bronze— raw landings on Cloud Storage as BigLake/Iceberg tables, queried in place (your 13:00 lab)
- Silver— cleansed, conformed tables built by Dataform incremental models, keys conformed across sources
- Gold— native BigQuery marts, partitioned and clustered with this morning's M9 levers
- Consumers— engineers work in Bronze and Silver; Power BI and business users read only Gold
- Trust— Bronze is received, Silver is verified, Gold is decision-grade

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Cymbal migration strategies

For a company, such as Cymbal, which might be running on a traditional on-premises data warehouse (such as Teradata or Hadoop), migrating to a cloud-native data lakehouse is a significant project.

A complete, one-time migration is often too risky and disruptive. Instead, a phased, use-case-driven approach is usually more successful.

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Migration strategy

01 Set up the core infrastructure on Google Cloud

02 Start with a high-impact use case

03 Migrate relevant data

04 Build the new pipelines and reports

05 Decommission and iterate



Their phased migration is your phased migration

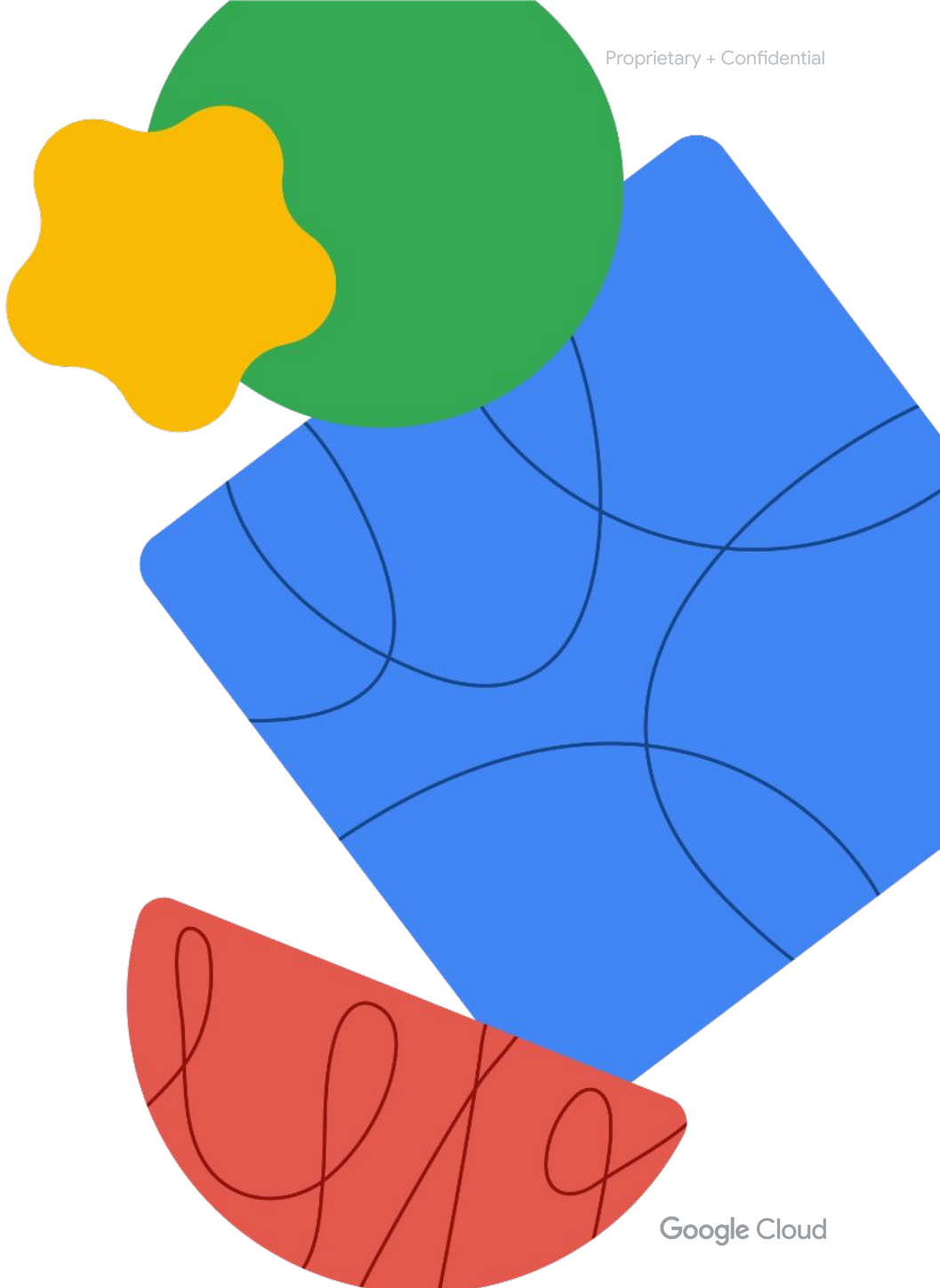
A one-time move off the warehouse is too risky — phase it by use case.

- Stand up the core first— projects, datasets, IAM, billing guardrails
- Pick one painful extract chain— Bronze to Dataform Silver to Gold mart to the existing Power BI report
- Parallel-run, then cut over— old job and new pipeline side by side until the numbers match
- Decommission last— retire the extract only after the Gold mart feeds the report
- Repeat per use case— this afternoon's capstone starts that exercise on paper

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Cost management and optimization best practices

A proactive approach to cost management.



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Best practices (1 of 2)

01

Choose the right storage class

Not all data needs to be accessed with the same frequency. Consider storage class in Cloud Storage.

02

Optimize BigQuery queries

Use features, such as partitioning and clustering, as well as writing efficient SQL queries.

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Best practices (2 of 2)

03

Use BigQuery's flat-rate pricing

Great for predictable workloads.

04

Set up budgets and alerts

In Google Cloud's billing tools.

M10 — What you now have

- Governance mapped to your single-source-of-truth requirement— catalog, PII protection, column and row security
- Your medallion design confirmed in Google's own teaching deck— Bronze, Silver, Gold, zone for zone
- A GCP home for each zone— GCS/BigLake Bronze, Dataform Silver-to-Gold, BigQuery Gold marts for Power BI
- A phased migration path and the cost levers— storage class, bytes scanned, budgets and alerts
- Next: M11— the BQML direction and the Power BI last mile on your Gold marts